- With costs capped at the 99th percentile the bill from 30 falls from £38,000 to £32,000 and uniform prevention from £35,000 to £22,000, so P/C falls to 0.69, 0.52 and 0.31 at 30, 50 and 70 on the AR(1) plus "spike" classes (the cap removes most of the sick tail, which is where prevention's pounds are); the within-class part of the bill is 11%, 9% and 8%; the between-class shares of P are +31%, +23% and +6%, and the class-targeting gains 90%, 28% and 47%.
- The one change is at 50, where the middle class rather than the worst has the highest return under the capped curve.

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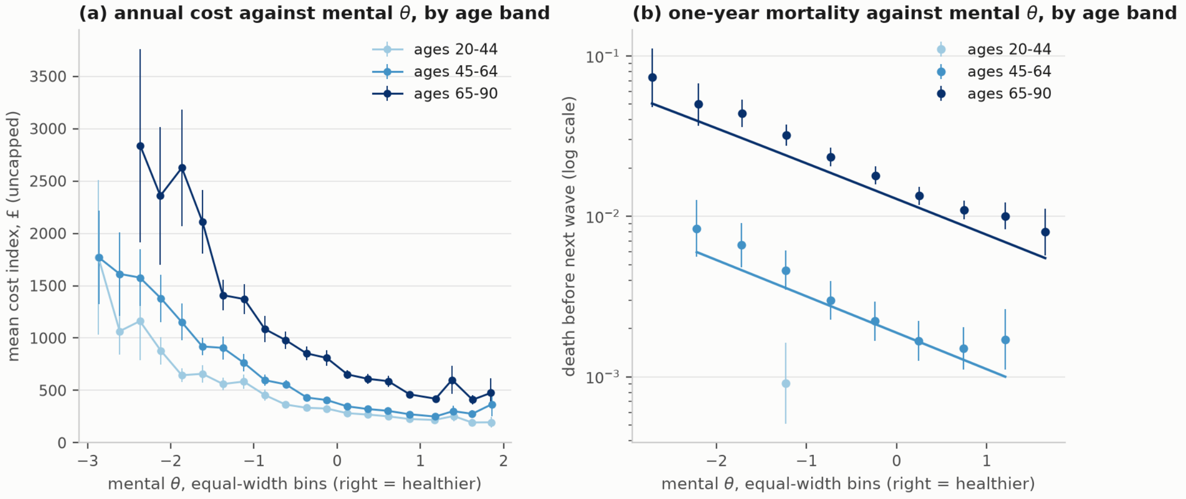
H Mental health

The mental GRM: the three SF-12 mental testlets and two GHQ-12 testlets, no depression diagnosis, the second channel of the multidimensional model of Section 4.3, here on its own. Higher is better; standardised on the pooled sample. The K-means and the mixtures use the multidimensional contract's rows, the health contract's people with a mental score on every row (38,181 people), so they share a sample with the multidimensional fits.

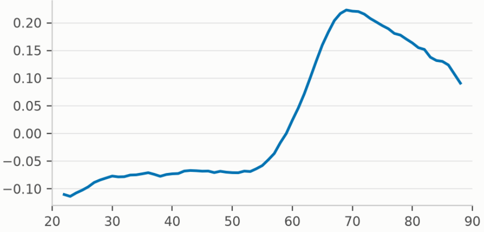


UKHLS, the mental GRM (three SF-12 mental testlets and two GHQ-12 testlets, no depression diagnosis), higher = better, ages 20-90. Cost: the cost index (in-patient, out-patient and GP), uncapped, waves 7-15, 209,077 person-waves; bins with at least 150 person-waves, 95% intervals. Mortality: death before the next wave (one year), waves 1-14, 351,769 person-waves at risk, 1,737 deaths; bins with at least 10 deaths, Wilson 95% intervals; lines are the within-band logit, linear in the measure and quadratic in age, drawn at the band's mean age. Measures standardised on the pooled sample, so a bin means the same health at every age.

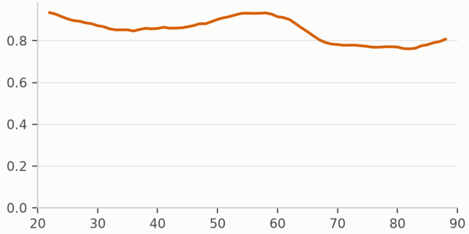
Figure H.43: As Figure 1, on the mental GRM.

- Cost is convex in mental health too: the pooled curvature has $t = 21.4$ against 36.9 on θ, and the slope is about half of θ's (£144 against £279 per sd at the mean). Unlike θ, the gradient steepens with age ($t = 22$ on the slope-by-age interaction, against -1.5 on θ): a low mental score costs little at 20–44 and a lot at 65–90.
- Mortality falls with mental health at every age, with a gradient of 0.57 log-odds per sd against 0.90 on θ, and it weakens with age ($t = -3.0$), where on θ it is stable.

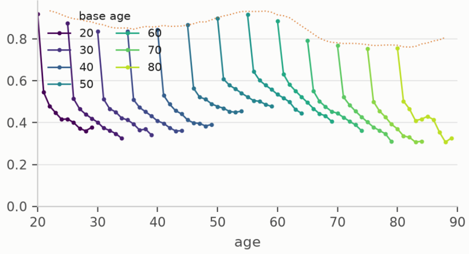
(a) mental θ (latent, pooled $N(0,1)$): mean



(b) variance



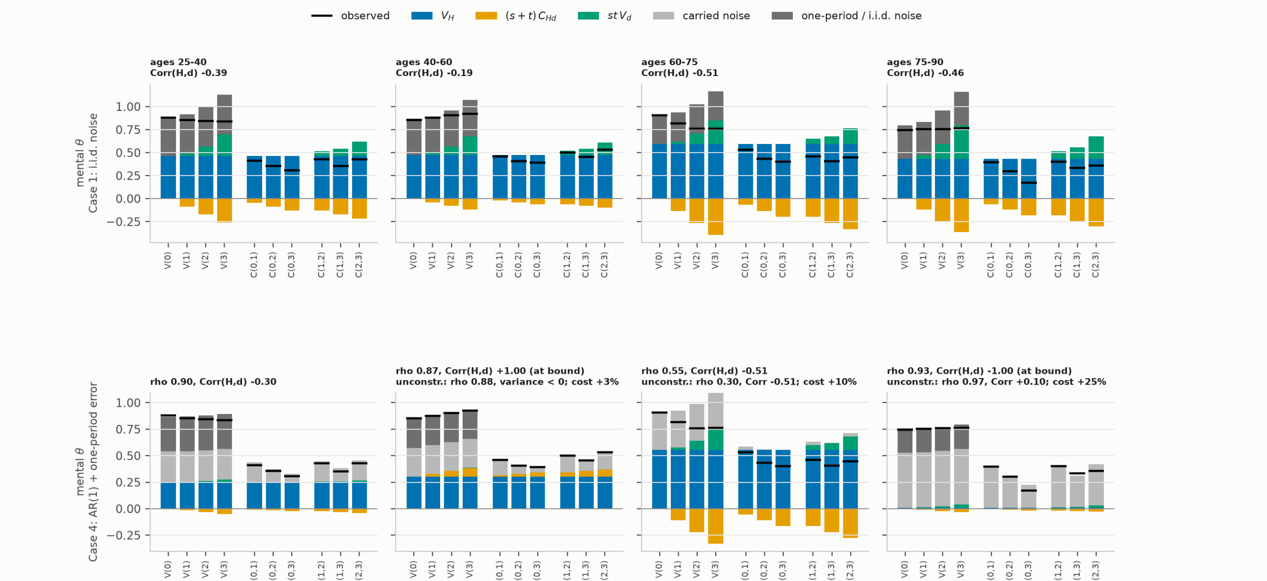
(c) rows of the covariance matrix



The mental GRM: three SF-12 mental testlets and two GHQ-12 testlets, no depression diagnosis; higher = better. 376,019 person-waves, 66,841 people, ages 20-90. Profiles are five-year centred rolling means; cells with fewer than 100 person-waves are dropped. Each covariance row starts on the diagonal at its five-year base-age bin (dotted: the variance profile) and runs nine years. Panels (b) and (c) share a vertical scale.

Figure H.44: As Figure 2, on the mental GRM: mean path, variance profile and covariance rows, model-free.

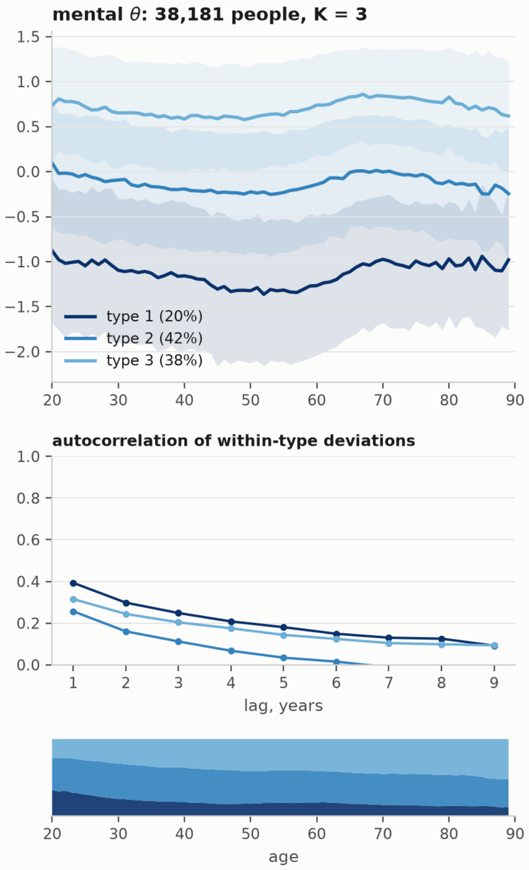
- The mean path is nothing like the physical one: flat at about −0.08 from 20 to 55, then up by 0.3 sd to a peak at about 70, then down again. The variance is flat at 0.85–0.93 across the whole age range, against the hump on θ.
- Roughly half of the variance is one-period: every covariance row drops from about 0.9 on the diagonal to 0.45–0.55 a year later, then decays slowly. On θ the first-year drop is about a fifth. This is the fast mental channel of the multidimensional fits, where its “spike” share is 0.34–0.55 by class.



The mental GRM: three SF-12 mental testlets and two GHQ-12 testlets, no depression diagnosis; higher = better. Four ages four years apart, pairwise moments pooled over base ages within each band; a base age enters when every cell rests on at least 50 people observed at both of its ages. Properly constrained fit: V_H, V_d and noise variances non-negative and |Corr(H,d)| <= 1 imposed in the estimation, rho profiled. Where the unconstrained fit was inadmissible the title gives it and the fit cost of the constraint in squared error. Where V_H falls below 5% of the first variance the level, and so Corr(H,d), is not identified. Bars stack the fitted layers; black ticks are the observed moments. V(k): the variance at the k-th age; C(k,l): the covariance between ages k and l.

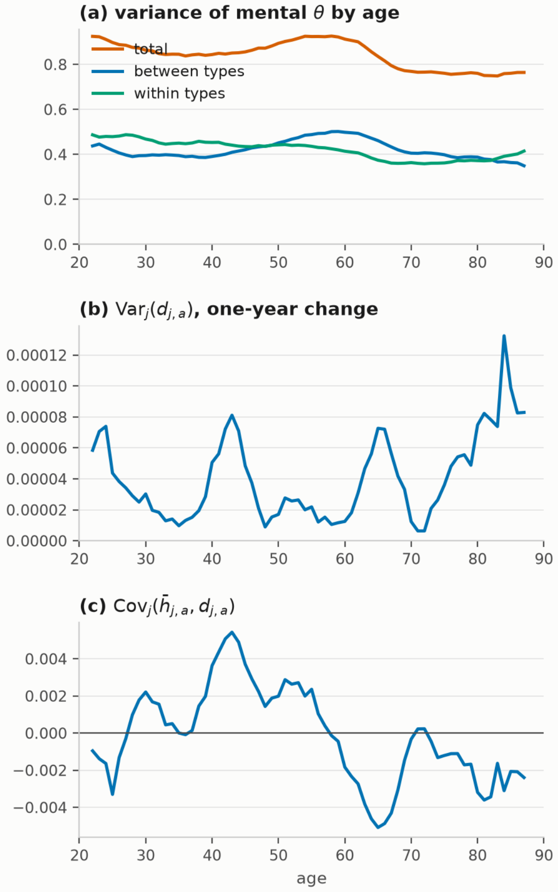
Figure H.45: As Figure 3, Cases 1 and 4 on the mental GRM.

- Under Case 1, Corr(H, d) is negative in every band (-0.39, -0.19, -0.51, -0.46): the shocks pull back towards the level throughout, with no positive-correlation band in mid-life as on θ. Under Case 4 the fit is not clean: -0.30 at 25–40 with $\rho = 0.90$, but at 40–60 and 75–90 the constraint binds ($+1$ and -1) and at 60–75 the persistence falls to 0.55. With half the variance one-period, the four-year moments leave little to identify the persistent part from.



The multidim health contract: 38,181 people with a mental score on every row,
at least three observed ages 20-89, one row per person-age; the mental GRM
(three SF-12 mental testlets and two GHQ-12 testlets, no depression diagnosis).
Partial K-means with 50 seeded starts,
types ordered worst to best. Top: cluster means where at least 25 members
are observed, shaded to +/- 1 within-cluster sd. Middle: correlation of a
person's deviation from their type's mean at age a with their deviation
k years later. Bottom: type composition of the person-waves observed at each age.

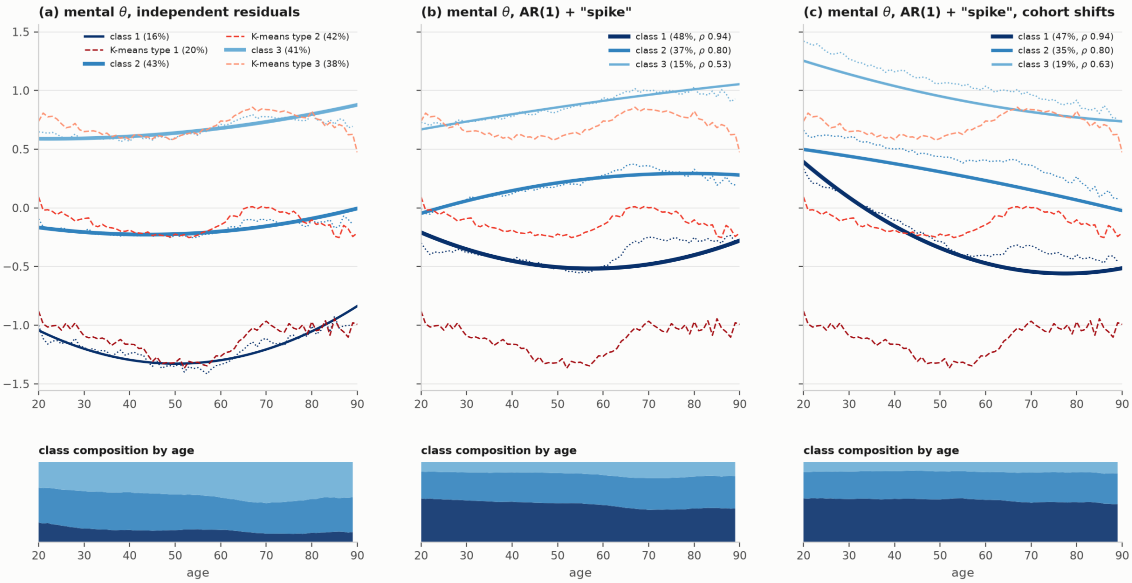
Figure H.46: As Figure 4, partial K-means types ($K = 3$) on the mental GRM, on the multidimensional contract's rows.



Row (a): the cross-sectional variance of the measure among the clustered people, split by the age-specific type composition; five-year rolling means. Rows (b) and (c): the smoothed type mean paths (five-year rolling means, ends extended linearly), d the one-year change, moments across types with person-level shares, then a further five-year rolling mean.

Figure H.47: As Figure 5, between versus within by age, on the mental GRM types.

- The type shares are almost those of θ (.20/.42/.38 against .18/.43/.40), but the paths are flat and parallel, at about -1.1, -0.15 and $+0.65$, each carrying the mid-life dip and the rise after 60 of the mean path; the gaps between the types hardly move with age.
- The types are different people. Agreement with the θ typology on the 38,181 shared people is ARI 0.14 with 58% in the same type (0.14 and 54% against h), against ARI 0.37–0.71 between the physical measures.
- The between-type share of the variance is 45–52% at every age, against a rise from 49% at 25 to 64–66% at 55–70 on θ. Within-type autocorrelation at one year is 0.26–0.39 against 0.37–0.64 on θ, and is near zero by six years for the middle type: the persistent part of mental health is smaller, consistent with the covariance rows of Figure H.44.
- The level-change covariance across types is positive from the late twenties to the late fifties and negative from about 58 to 70, then small: the types fan out slowly in mid-life and close up around retirement, when the mean rises.



K = 3 quadratic growth mixtures on the mental GRM on the multidim health contract (38,181 people with a mental score on every row, 325,822 person-ages), parameters at the posterior mean; class 1 is worst health. Left: independent residuals; middle: an AR(1) latent state plus a one-period "spike", with the class persistence in the legend; right: the same with one level shift per birth decade, common to the classes, paths and observed means at the 1950s, every score net of its decade shift. Line width is proportional to the class share. Dotted: the observed class mean at each age, the measure averaged over the people observed there weighted by their posterior class probabilities. Dashed red: the partial K-means type means on the same rows. Bottom: the posterior class composition of the person-waves observed at each age.

Figure H.48: As Figure 6, $K = 3$ mixtures on the mental GRM, on the multidimensional contract's rows: (a) independent residuals, (b) an AR(1) latent state plus a one-period "spike", (c) the same with one level shift per birth decade common to the classes, paths and observed class means at the 1950s with every score net of its decade shift. Dashed red: the K-means types of Figure H.46.

- Under independent residuals the mixture reproduces the K-means types: shares .16/.43/.41 against .20/.42/.38, class paths at about -1.3, -0.2 and $+0.6$ at 50, each with the dip to 50 and the rise after 60.
- Under AR(1) plus "spike" the classes are a different partition. The persistence is .94/.80/.53 with a "spike" scale of 0.54, so the signal shares are 0.72/0.45/0.29, and the largest class (48%) is the lowest, running from -0.2 at 20 to -0.5 at 55 and back; nothing sits near the K-means worst type at -1. With half of the variance one-period, the mixture separates people by how persistent their mental score is rather than by its level: the small top class (15%) is mostly "spike". The class composition is flat in age.
- With birth-decade shifts the shares and persistence barely move (.47/.35/.19; .94/.80/.63), but the shifts are large: relative to the 1950s, $+0.2$ for the 1920s–40s and -0.9 for the 2000s, 1.1 sd across the 1940s–2000s, twice the physical shift. At the 1950s every class path then declines with age, so the rise after 60 in Figure H.44 is being read as the older cohorts' higher level. The age–cohort caveat of Section 4.3 applies with more force here, since the whole post-60 rise sits inside the fifteen-year window where age and birth year are nearly collinear.